THE VIRTUAL MEET EXPERIENCE

2020 - 2021

HS VIRTUAL CHALLENGE MEET #1

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2020-2021
HS Virtual Challenge Meet #1
Computer Science Test

General Directions:

1) DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
2) **NO CALCULATORS of any kind may be used.**
3) You have 45 minutes to complete this contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
4) Papers may not be turned in until forty-five minutes have elapsed. If you finish the test before the end of the allotted time, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
5) All answers must be written on the answer sheet/Scantron card provided. Indicate your answers in the appropriate blanks provided on the answer sheet or on the Scantron card. Clean erasures are necessary for accurate Scantron grading.
6) You may place as many notations as you desire anywhere on the test paper except on the answer sheet or Scantron card which is reserved for answers only.
7) You may use additional scratch paper provided by the contest director.
8) All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers. **All provided code segments are intended to be syntactically correct, unless otherwise stated (i.e. `error` is an answer choice). Ignore any typographical errors and assume any undefined variables are defined as used.**
9) A reference to commonly used Java classes is provided with the test and you may use this reference during the contest. You may detach the reference sheets from the test booklet but DO NOT DO SO UNTIL THE CONTEST BEGINS.
10) Assume that any necessary import statements for Standard Java 12 Packages and classes (e.g. .lang, .util, System, Math, Double, etc.) are included in any programs or code segments that refer to methods from these classes and/or packages.

Scoring:

1) All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for each incorrect answer.

For Computer Science practice tests and hands-on materials, go to www.apluscompsci.com

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Note: Correct responses are based on **Java SE Development Kit 12 (JDK 12)**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java SE 12 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported using:** ***import static java.lang.System.*;***

QUESTION 1

11011_2 plus 10101_2 = ?

A. 299_4 B. 119_6 C. 61_8 D. 47_{10} E. 40_{12}

QUESTION 2

What is output by the code to the right?

A. 7 B. 3
C. 2 D. 1
E. 0

```
int a = 7 % 3;
out.println( a );
```

QUESTION 3

What is output by the code to the right?

A. 378 B. 18
C. 108 D. 315
E. 37""8

```
out.println(3 + 7 + "" + 8);
```

QUESTION 4

What is output by the code to the right?

A. 12 B. l1 C. -1 D. 14 E. 13

```
String best = "aplusccompsci.com";
out.println( best.indexOf('.') );
```

QUESTION 5

What is output by the code to the right?

A. true B. false C. sam D. && E. not

```
boolean sam = false;
out.println( sam && true );
```

QUESTION 6

What is output by the code to the right?

A. 4.6 B. 4.7 C. 4.0 D. 5.0 E. 5

```
out.println( Math.ceil( 4.6 ) );
```

QUESTION 7

What is output by the code to the right?

A. 4 B. 4.0 C. 4.5 D. 5 E. 5.0

```
Double b = (double)9 / 2;
out.println( b );
```

QUESTION 8

What is output by the code to the right?

A. 123 B. 12 C. 23 D. 13 E. 3

```
int cnt = 10;
if( cnt > 5 )
  out.print( 1 );
if( cnt > 10 )
  out.print( 2 );
out.print( 3 );
```

QUESTION 9

What is output by the code to the right?

A. 12345 B. 0246 C. 024 D. 02 E. 0

```
for( int i=0; i<6; i+=2 )
  out.print( i );
```

QUESTION 10

What is output by the code to the right?

A. goif
B. go
C. goifat
D. if
E. atgo

```
String [] s = {"if", "go", "at"};
out.print( s[1] );
out.print( s[0] );
```

QUESTION 11

Which of the following could fill **<blank>** in the code to the right?

A. int x
B. double x
C. long x
D. float x
E. All of the above

```
Scanner s = new Scanner( System.in );
<blank> = s.nextInt();
```

QUESTION 12

What is output by the code to the right?

A. 22
B. 20
C. 18
D. 15
E. Nothing is returned due to a syntax error.

```
int sum = 0;
for( int i=0; i<9; i+=2 )
  sum += i;
out.print( sum );
```

QUESTION 13

What is output by the code to the right?

A. 9.0
B. 14.0
C. 10.0
D. 6.0
E. 7.0

```
double w = 11 % 2 + 3 * 4 / 2;
out.print( w );
```

QUESTION 14

What is output by the code to the right?

A. 22 B. 11 C. 9 D. 16 E. 10

```
int a = 9 | 22 & 4 ^ 7;
out.print( a );
```

QUESTION 15

What is output by the code to the right?

A. []
B. [3]
C. [6,3]
D. [3,6]
E. There is no output due to an error.

```
List< Short > m;
m = new ArrayList< Short >();
m.add( 3 );
m.add( 1, 6 );
out.print( m );
```

QUESTION 16

What is output by the code to the right?

A. null
B. 0.00
C. nu
D. nul
E. There is no output due to a runtime error.

```
Double d = null;
System.out.printf("%.2f", d);
```

QUESTION 17

What is output by the code to the right?

A. 5 B. 1 C. 0 D. 2 E. 6

```
System.out.println( 1 % 5 );
```

QUESTION 18

What is output by the code to the right?

A. 11 B. 14 C. 19
D. 27 E. 20

```
int sum=0;
for(int j=0; j < 6; j++){
  for(int k=0; k < j; k++){
    sum+=k;
  }
  if(j%2==1)
     sum-=3;
}
out.println(sum);
```

QUESTION 19

How many times does this loop iterate?

A. 7 B. 8 C. 6 D. 9 E. 5

```
int z = 07;
do{
  z = z - 1;
}while(z > 1);
```

QUESTION 20

What is output by the code to the right?

A. 25.0 B. 36.0 C. 25 D. 36
E. There is no output due to a syntax error.

```
float x;
x = Math.pow(Math.round(5.5),2);
out.println(x);
```

QUESTION 21

Which of the following reserved words is used to access a member of a parent class?

A. extends B. implements C. parent D. super E. try

QUESTION 22

Which of the following operators divides by 2?

A. >> B. << C. & D. | E. ^

QUESTION 23

Which of the following could fill **<*1>**?

A. Char B. Bool C. Byte
D. Int E. more than one of these

```
Set< <*1> > set;
set = new TreeSet< <*1> >();
```

QUESTION 24

What is output by the code to the right?

A. 5 B. 10 C. 3 D. 3.0 E. 3.33

```
int[] ray;
ray = new int[4];
ray[0]=5;
ray[1]=ray[0]*2;
ray[2]=ray[1]/3;
out.println(ray[2]);
```

QUESTION 25

What is returned by the call `whoot(7)`?

A. 5040 B. 720 C. 720.0 D. 24 E. 5040.0

```
static double whoot(double x){
   double ans=1.0;
   for(double r=x; r>=1; r--)
      ans=ans*r;
   return ans;
}
```

QUESTION 26

Which of the following could fill **<*1>**?

A. `public`
B. `static`
C. `final`
D. A and B only
E. A, B, and C

```
public interface Inter{
   <*1> int x = 1024;
}
```

QUESTION 27

What is output by the line marked **//1**?

A. `[T, P, 0, P]`
B. `[P, 0, P, T]`
C. `[T, I, P, 0, P]`
D. `[0, P, P]`
E. `[P, 0, P, I, T]`

QUESTION 28

What is output by the line marked **//2**?

A. `[I, P, 0, P]`
B. `[P, 0, P, T]`
C. `[P, 0, P, I]`
D. `[0, P, P]`
E. `[P, 0, P]`

```
Stack<Character> s;
s = new Stack<Character>();
s.add('P');
s.add('0');
s.add('P');
s.add('I');
s.add('T');
out.println(s);              //1
out.println(s.pop());
out.println(s);              //2
```

QUESTION 29

What is output by the code to the right?

A. `64`
B. `32`
C. `256`
D. `128`
E. `64`

```
out.println(4096 >> 7);
```

QUESTION 30

What is output by the code to the right?

A. `true 0`
B. `false 0`
C. `null 0`
D. `false 1`
E. There is no output due to syntax error.

```
int[] a = new int[7];
boolean[] b = new boolean[7];
for(int i = 0; i < 7; i++){
   a[i] = a[i] + 1;
}
System.out.println(b[5] +" "+ a[5]);
```

QUESTION 31

What is output by the line marked **//1**?

A. `$`
B. `7`
C. `W`
D. `6`
E. `5`

QUESTION 32

What is output by the line marked **//2**?

A. `$`
B. `7`
C. `W`
D. `6`
E. `5`

```
ArrayList<Character> cList;
cList = new ArrayList<Character>();
cList.add('$');
cList.add('&');
cList.add('7');
cList.add('4');
cList.add('W');
ListIterator iter;
iter = cList.listIterator();
iter.next();
iter.next();
iter.remove();
iter.add('6');
iter.previous();
out.println(iter.next());      //1
iter.set('5');
iter.next();
iter.previous();
iter.previous();
out.println(iter.next());      //2
```

QUESTION 33

What is output by the code to the right?

A. 23

B. 21

C. 20

D. 25

E. 19

```
int trick = 19;
trick++;
trick++;
++trick;
trick=trick++;
trick=trick++;
out.println(++trick);
```

QUESTION 34

Which algorithm best describes method `what`?

A. binary search

B. quick sort

C. linear search

D. selection sort

E. merge sort

QUESTION 35

What is the best case run time for method `what`?

A. O(N)

B. O(N/2)

C. O(1)

D. $O(\log_2 N)$

E. $O(N^2)$

```
static int what(double[] d, double y)
{
   int spot = -1, x = 0;
   while(x < d.length)
   {
      if(d[x++] == y)
        spot = x;
   }
   return spot;
}
```

QUESTION 36

What is output by the code to the right?

A. 26 7

B. 25 8

C. 30 9

D. 25 5

E. 26 5

```
int s = Short.MAX_VALUE, x = s;
int y = 0;

for(s = 0; x > 0; y++)
{
  s += x % 10;
  x /= 10;
}

System.out.println(s + " " + y);
```

QUESTION 37

What is the in-order traversal of the binary tree created when the following values are inserted in order?

```
17 4 5 29 42 3 90 9 100 8
```

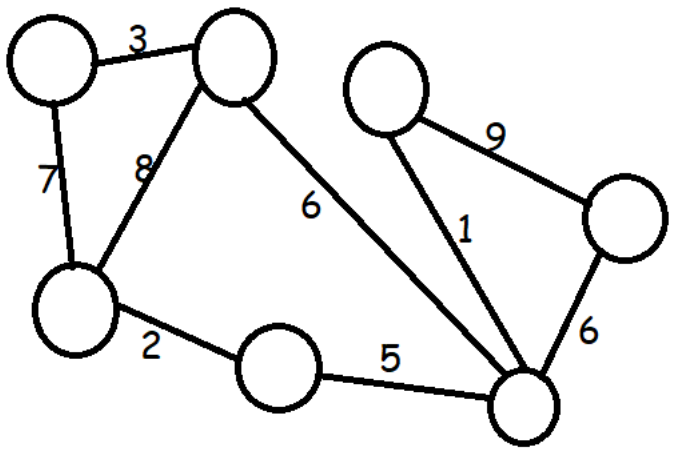
QUESTION 38

Fully simplify the following expression.

```
(A^(B*C))*(A*C)
```

QUESTION 39

What is the cost of the most expensive cycle in the following graph?



QUESTION 40

What is the most simplified Boolean equivalent to the following truth table?

A	B	Result
1	1	1
1	0	1
0	1	0
0	0	0

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets™

Computer Science - Student Answer Sheet

1) _____
2) _____
3) _____
4) _____
5) _____
6) _____
7) _____
8) _____
9) _____
10) _____

11) _____
12) _____
13) _____
14) _____
15) _____
16) _____
17) _____
18) _____
19) _____
20) _____

21) _____
22) _____
23) _____
24) _____
25) _____
26) _____
27) _____
28) _____
29) _____
30) _____

31) _____
32) _____
33) _____
34) _____
35) _____
36) _____
37) _____
38) _____
39) _____
40) _____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____